BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: 4465

Retyped faithfully from the original board paper — same questions, no changes.

Q. Paper : II (Objective Type)

سوالیہ پرچہ: II (معروضی طرز)

QQ. 1. 1221 — (Inter Part - II) (Session 2017-19 to 2019-21) — Biology (Objective) — Paper (II)

Q. Time Allowed:- 20 minutes — PAPER CODE 4465 — Maximum Marks:- 17

Q1. Coniferous forests located at high altitude are called

(A) (B)

(C) (D)

Q2. CFCs are produced by

(A) (B)

(C) (D)

Q3. Which organ is the central station of metabolism

(A) (B)

(C) (D)

Q4. Most invertebrates, fish, amphibians and reptiles are included in

(A) (B)

(C) (D)

Q5. The end of muscle which remains fixed when muscle contracts

(A) (B)

(C) (D)

Q6. Muscle is connected to bone by

(A) (B)

(C) (D)

Q7. The processes conducting impulses away from the cell body are called

(A) (B)

(C) (D)

Q8. Human sperms are formed in

(A) (B)

(C) (D)

Q9. Discharge of egg from the ovary is called

(A) (B)

(C) (D)

Q10. The unspecialized cells present in flatworms and planaria are

(A) (B)

(C) (D)

Q11. Which one of the following is Initiation Codon

(A) (B)

(C) (D)

Q12. Synapsis takes place in

(A) (B)

(C) (D)

Q13. Failure of separation of chromosomes is called

(A) (B)

(C) (D)

Q14. Secretors have dominant Secretor gene "Se" on chromosome

(A) (B)

(C) (D)

Q15. Urine is a preferable vehicle for Biotechnology product than

(A) (B)

(C) (D)

Q16. The change in frequency of alleles at a locus that occurs by chance

(A) (B)

(C) (D)

Q17. The relationship between insects and flowering plants is an example of

(A) (B)

(C) (D)

Board typos kept verbatim: 'Coniferous forests located at high altitude are called', 'CFCs are produced by', 'Foam', 'Industrial machines', 'Air conditioners and Refrigerators', 'Aeroplanes', 'Which organ is the central station of metabolism', 'Most invertebrates, fish, amphibians and reptiles are included in', 'Ectotherm', 'Endotherm', 'Heterotherm', 'Homeotherm', 'The end of muscle which remains fixed when muscle contracts', 'Insertion', 'Belly', 'Origin', 'Tendon', 'Muscle is connected to bone by', 'Ligament', 'Z-line', 'The processes conducting impulses away from the cell body are called', 'Dendron', 'Dendrites', 'Nissl granules', 'Axon', 'Human sperms are formed in', 'Seminiferous tubules', 'Epididymis', 'Vasa efferentia', 'Ureter', 'Discharge of egg from the ovary is called', 'Oogenesis', 'Menstruation', 'Ovulation', 'Menopause', 'The unspecialized cells present in flatworms and planaria are', 'Neoblast', 'Osteoblast', 'Epiblast', 'Hypoblast', 'Which one of the following is Initiation Codon', 'GUA', 'UGA', 'AUG', 'GAU', 'Synapsis takes place in', 'Leptotene', 'Zygotene', 'Pachytene', 'Diakinesis', 'Failure of separation of chromosomes is called', 'Disjunction', 'Separation', 'Non-disjunction', 'Metastasis', 'Secretors have dominant Secretor gene "Se" on chromosome', 'Urine is a preferable vehicle for Biotechnology product than', 'Blood', 'Milk', 'Plasma', 'Serum', 'The change in frequency of alleles at a locus that occurs by chance', 'Crossing over', 'Mutation', 'Migration', 'Genetic Drift', 'The relationship between insects and flowering plants is an example of', 'Parasitism', 'Predation', 'Mutualism', 'Commensalism'.

Paper : II (Essay Type)

پرچہ: II (انشائیہ طرز)

QSection I. 1221 — (Inter Part - II) (Session 2017-19 to 2019-21) — Biology (Subjective) — Paper (II)

Q. Time Allowed: 2.40 hours — Maximum Marks: 68

QSection II. Section II

Note: Attempt any three questions. (8 × 3 = 24)

Board typos kept verbatim: 'Compare physical control system with living control system.', 'Compare hydrophytes with mesophytes.', 'Describe thermostat function and feed back mechanism in human.', 'How hematoma is formed?', 'Differentiate between fibers and sclerides.', 'Differentiate between hyaline cartilage and fibrocartilage.', 'How seedless fruits are formed?', 'Define fruit set in plants.', 'Differentiate between phytoplanktons and zooplanktons.', 'Write human impact on coniferous alpine and boreal forests.', 'Differentiate

between renewable and non renewable resources.', 'What are green house and green house gasses?', 'Differentiate between stimulus and respons.', 'What is epilepsy?', 'Define Kinesis with an example.', 'Differentiate between complete dominance and co-dominance.', 'Define population and gene pool.', 'Define linkage and crossing over.', 'Give two possible ways to get gene of intrest.', 'What is gene therapy?', 'Differentiate between transgenic plants and transgenic animals.', 'What is symbiosis name its types.', 'Differentiate between ecosystem and biosphere.', 'Define nitrification and denitrification.', 'What are apical meristems?', 'Explain the compensatory effect in growth.', 'What are Purines and Pyrimidines?', 'What is Anti-Sense strand?', 'How the Genetic code is universal?', 'Differentiate between interphase and mitotic phase.', 'What is metastasis?', 'What is membrane Invagination Hypothesis?', 'Write a note on the Vestigial organs.', 'Discuss osmoregulation in plants.', 'Explain different stages of nitrogen cycle with the help of a diagram.', 'Give an account of bones of Skull (cranial and facial)', 'Define chromosomes. Give their chemical composition in detail.', 'Differentiate between axons and dendrites.', 'What is acid rain?', 'Define parthenogenesis. Explain different types of parthenogenesis.', 'Describe the genetics of colour-blindness in humans.', 'Discuss role of Nucleus in development through an experiment.', 'Explain main points developed by Darwin in his book "The Origin of Species".', 'Answer briefly any Eight parts from the followings:-', 'Answer briefly any Six parts from the followings:-', 'Note: Attempt any three questions.', 'feed back', 'intrest', 'gasses', 'respons', 'Kinesis', 'hematoma', 'phytoplanktons', 'zooplanktons', 'parthenogenesis', 'colour-blindness', 'Invagination Hypothesis', 'Vestigial organs'.